Case Records of the Wexford General Hospital

A Clinicopathological Exercise • Synthetic Teaching Case

Case 22-2026: A 39-Year-Old Man with Episodic Headache, Palpitations, and Paroxysmal Hypertension

The following is an entirely fictional teaching case. All names, dates, case numbers, institutions, laboratory values, and narrative details are invented for instructional purposes and do not depict any real patient, physician, or institution.

Presentation of the Case

Dr. Priya Ramachandran (Medicine): A 39-year-old man was admitted to this hospital because of six months of recurrent paroxysmal symptoms and an episode of severe hypertension.

The patient had been well until approximately six months before this admission, when he began to experience discrete, stereotyped "spells." Each episode began abruptly, within seconds, with a pounding, bilateral, holocephalic headache that he described as a pressure building behind both eyes. It was accompanied almost simultaneously by forceful palpitations, which he perceived as a hammering in his chest and neck, by profuse drenching sweating, and by a fine tremor of the hands. During the episodes his skin, particularly of the face and hands, turned strikingly pale and cold rather than flushed, a feature his wife corroborated. The spells were attended by a profound sense of apprehension and impending doom, together with nausea and, at times, breathlessness. Each lasted ten to thirty minutes and then subsided gradually, leaving him fatigued and occasionally with polyuria. Between episodes he felt nearly, though not entirely, well.

At the onset the spells occurred once or twice weekly, but over the ensuing months they increased to several times per week. The patient came to recognize certain provocations: episodes were sometimes precipitated by physical exertion such as climbing stairs or lifting boxes, by bending forward or straining, and, on more than one occasion, by micturition. Others arose without any discernible trigger, including several that woke him from sleep. He had identified no consistent relationship to meals or caffeine, and curtailing coffee and exercise had not helped.

Approximately four months before admission, the patient sought care from his primary physician. On the basis of the episodic symptoms, the prominent palpitations, the tremor, and the overwhelming sense of dread, a diagnosis of panic disorder was made, and sertraline was prescribed and uptitrated. It produced no reduction in the spells, and the patient privately doubted the diagnosis, because the episodes seemed to arise from his body rather than his thoughts and were not relieved by reassurance or breathing techniques.

Over the same interval, the patient noted an unintentional loss of approximately 8 kg despite a preserved, if not increased, appetite. He also described new intermittent constipation, and during the spells he felt uncomfortably hot and intolerant of warmth, though he was not aware of fever.

Two weeks before admission, while at work, the patient had an unusually severe and prolonged spell. A colleague drove him to a local emergency department, where the blood pressure was recorded as 224/126 mm Hg with a regular tachycardia at 132 beats per minute. He was treated with intravenous antihypertensive medication, and over the following hour his blood pressure and heart rate fell as the symptoms abated. He was observed overnight and discharged. This episode and the documented hypertensive surge prompted his referral to this hospital and subsequent admission.

The patient's medical history was notable for hypertension, diagnosed two years earlier during an unrelated office visit and difficult to control. His blood pressure had remained elevated despite the sequential addition of amlodipine and lisinopril, and his physician had remarked that his pressures were unusually labile, some readings near normal and others markedly elevated. Three months before admission, routine testing had revealed a new fasting glucose in the diabetic range; he had no polydipsia or

polyuria apart from the post-episode polyuria already described. He had undergone no surgical procedures. His medications were amlodipine, lisinopril, and sertraline; he had no known drug allergies, and his immunizations were up to date.

The patient's family history was known to him only in outline but was of particular interest. His father had long-standing hypertension and, in his fifties, had undergone an operation for what the patient understood to be a "tumor of the thyroid gland"; he recalled that his father took thyroid hormone afterward and was monitored with blood tests. A paternal uncle had reportedly had a tumor removed "from near one kidney" at a relatively young age, though details were unavailable. There was no known family history of a defined genetic syndrome, sudden cardiac death, or parathyroid disorders. His mother and two siblings were well.

The patient worked as an accountant and lived with his wife and two children, with no occupational exposures of note. He had never smoked or used electronic cigarettes, drank alcohol rarely, and had never used illicit drugs or stimulants of any kind; he specifically denied cocaine, amphetamines, and over-the-counter decongestants and weight-loss preparations. He had not traveled abroad and had no unusual dietary practices.

On review of systems, the patient described, during his spells, the episodic pallor, sweating, tremor, palpitations, headache, nausea, and heat intolerance already noted; he denied diarrhea, wheezing, or cutaneous flushing or hives. Between episodes he reported the new constipation and the weight loss. He denied chest pain at rest, exertional angina, orthopnea, paroxysmal nocturnal dyspnea, syncope, visual disturbance, focal weakness, polydipsia, kidney stones, bone pain, or change in his voice or swallowing. He had noticed no neck mass.

On examination between episodes, the patient was thin and appeared anxious but was in no acute distress. The supine blood pressure was 150/95 mm Hg and the heart rate 96 beats per minute and regular; on standing, the blood pressure fell to 128/82 mm Hg and the heart rate rose to 112 beats per minute, with momentary lightheadedness. The temperature was 36.8°C, the respiratory rate 16 breaths per minute, and the oxygen saturation 98% while he was breathing ambient air. The body-mass index was 21.4. The hands exhibited a fine resting and postural tremor, and the palms were cool and slightly moist. There was no exophthalmos or lid lag. The neck was supple; the thyroid gland was normal in size and contour, without a palpable nodule or bruit, and there was no cervical lymphadenopathy or mucosal neuroma of the lips or tongue. Funduscopy revealed arteriolar narrowing and arteriovenous nicking consistent with chronic hypertensive retinopathy, without papilledema. The jugular venous pressure was normal. The cardiac apical impulse was forceful and mildly displaced, with a regular rhythm, a fourth heart sound, and no murmur. The lungs were clear. The abdomen was soft and nontender, without organomegaly, palpable mass, or audible aortic or renal bruit. The extremities were without edema, and the peripheral pulses were full and symmetric. The skin showed no café-au-lait macules, axillary or inguinal freckling, or neurofibromas, and there were no mucocutaneous lentigines. Neurologic examination was nonfocal apart from the tremor; the patient was alert, fully oriented, and appropriately interactive.

The patient was admitted for monitoring and evaluation. During the first two hospital days he had two witnessed spells, each beginning abruptly with pallor, headache, and palpitations; during one, the blood pressure rose to 210/118 mm Hg with sinus tachycardia at 138 beats per minute and resolved over about twenty minutes, after which it returned to the 140s/90s. Laboratory studies obtained on admission and during the evaluation are shown in **Table 1** and **Table 2**.

Table 1. Hematologic, Chemical, and Metabolic Studies.

Variable	Patient's Value	Reference Range
Hemoglobin (g/dL)	17.4	13.5–17.5
Hematocrit (%)	52	41–53
White-cell count (×10⁹/L)	8.2	4.5–11.0
Platelet count (×10⁹/L)	271	150–400
Sodium (mmol/L)	141	135–145
Potassium (mmol/L)	4.3	3.5–5.0

Variable	Patient's Value	Reference Range
Chloride (mmol/L)	103	98–107
Bicarbonate (mmol/L)	25	22–30
Urea nitrogen (mg/dL)	16	7–20
Creatinine (mg/dL)	0.9	0.6–1.2
Calcium (mg/dL)	9.5	8.5–10.2
Phosphorus (mg/dL)	3.4	2.5–4.5
Fasting glucose (mg/dL)	168	70–99
Glycated hemoglobin (%)	6.6	<5.7
Aspartate aminotransferase (U/L)	24	10–40
Alanine aminotransferase (U/L)	27	7–35
Total cholesterol (mg/dL)	192	<200

Table 2. Endocrine and Catecholamine Studies.

Variable	Patient's Value	Reference Range
Plasma free metanephrine (nmol/L)	2.8	<0.50
Plasma free normetanephrine (nmol/L)	9.6	<0.90
24-hour urinary fractionated metanephrine (µg/24 h)	1,420	<320
24-hour urinary fractionated normetanephrine (µg/24 h)	5,980	<390
24-hour urinary norepinephrine (µg/24 h)	412	<80
24-hour urinary epinephrine (µg/24 h)	58	<20
Chromogranin A (ng/mL)	340	<93
Thyrotropin (µIU/mL)	1.8	0.4–4.0
Free thyroxine (ng/dL)	1.1	0.8–1.8
Plasma aldosterone (ng/dL)	9	≤21
Plasma renin activity (ng/mL/hr)	1.4	0.6–4.3
Aldosterone-to-renin ratio	6.4	<20
Morning serum cortisol (µg/dL)	13	5–25
Cortisol after 1-mg overnight dexamethasone (µg/dL)	1.1	<1.8

The plasma metanephrine measurements were obtained through an indwelling cannula after the patient had been supine and at rest for thirty minutes, fasting and having abstained from caffeine, with recognized interfering agents excluded. Urinalysis showed a specific gravity of 1.018 and was otherwise unremarkable, without protein, glucose, blood, leukocyte esterase, nitrites, or casts. An electrocardiogram obtained during a quiescent period showed sinus rhythm at 92 beats per minute with voltage criteria for left ventricular hypertrophy and repolarization abnormality; a tracing captured during a spell showed sinus tachycardia at 136 beats per minute without ischemic ST-segment shift.

Dr. Marcus Hale (Radiology): Computed tomography of the abdomen and pelvis was performed before and after the administration of intravenous contrast material, with a dedicated adrenal protocol that included unenhanced, portal-venous, and 15-minute delayed acquisitions. There was a well-defined, rounded mass arising from the right adrenal gland, measuring 4.8 cm in greatest dimension. On the unenhanced images the solid portions of the lesion measured approximately 38 Hounsfield units, well above the threshold of 10 units that characterizes a lipid-rich adenoma and indeed above 20 units. The mass enhanced avidly and heterogeneously after contrast administration, and there were internal areas of low attenuation consistent with cystic change and a focus of higher attenuation suggesting prior hemorrhage. On the delayed images the absolute contrast washout was approximately 34% and the relative

washout approximately 22%, both well below the values (≥60% and ≥40%, respectively) expected of a benign lipid-poor adenoma; the washout pattern was therefore atypical for an adenoma. The left adrenal gland was normal. Both kidneys were normal in size, position, and enhancement, without a renal mass or renal-artery abnormality, and there was no retroperitoneal or para-aortic lymphadenopathy and no hepatic lesion.

Magnetic resonance imaging of the adrenal glands was then performed. The right adrenal mass was isointense to liver on T1-weighted images and showed no loss of signal intensity on out-of-phase chemical-shift imaging, indicating the absence of intracytoplasmic lipid and again arguing against a lipid-rich adenoma. On T2-weighted images the lesion was markedly and heterogeneously hyperintense, approaching the signal intensity of cerebrospinal fluid—the so-called "light-bulb" appearance—with internal cystic and hemorrhagic foci. After gadolinium administration there was avid, heterogeneous enhancement.

Functional imaging was obtained with iodine-123 metaiodobenzylguanidine (^{123}I-MIBG) scintigraphy following appropriate thyroidal blockade. There was intense radiotracer accumulation localized to the right adrenal mass, with physiologic distribution elsewhere and no abnormal uptake in the contralateral adrenal gland, the para-aortic and paravertebral regions, the neck, the chest, the urinary bladder wall, or the skeleton, suggesting a solitary functioning lesion without evidence of extra-adrenal disease or metastasis.

Dr. Priya Ramachandran (Medicine): After the biochemical and imaging evaluation was complete, the patient underwent a period of pharmacologic preparation, and a diagnostic procedure was performed.

Differential Diagnosis

Dr. Eleanor Whitfield (Endocrinology): The clinical history, laboratory data, and imaging define a syndrome in which the history narrows the field, the biochemistry establishes the diagnosis, and imaging localizes the source.

The clinical problem can be named precisely. This is a previously healthy 39-year-old man with three intertwined abnormalities. First, he has *paroxysmal symptoms* in a remarkably stereotyped pattern: abrupt, self-limited spells of pounding headache, forceful palpitations, profuse diaphoresis, pallor, tremor, and a sense of doom, lasting tens of minutes, occurring several times weekly, and at times provoked by exertion, bending, straining, or micturition. Second, he has *hypertension* that is both sustained and paroxysmal—a young man with resistant hypertension on two agents, with strikingly labile pressures, and with a documented crisis of 224/126 mm Hg during a spell. Third, he has a *metabolic and constitutional* derangement: new hyperglycemia in the diabetic range, unintentional weight loss of 8 kg despite a preserved appetite, an upper-normal-to-high hematocrit, and an orthostatic fall in blood pressure on standing. Layered upon all of this is a *family history* of a paternal thyroid tumor and a relative with a tumor "near the kidney."

The differential diagnosis requires that the symptoms be considered at face value before a unifying endocrine diagnosis is accepted. The alternatives should be considered systematically, because each has been entertained in patients with this presentation.

A psychiatric disorder: panic and anxiety

This patient was diagnosed with panic disorder and treated with a selective serotonin-reuptake inhibitor, and the hypothesis merits consideration because the overlap is genuine. Panic attacks produce abrupt palpitations, sweating, tremor, dyspnea, chest tightness, and an overwhelming sense of dread, and they are far more common than any endocrine tumor. But the diagnosis of panic disorder here fails on several specific grounds. Panic attacks do not, as a rule, generate sustained, reproducible, severe hypertension; a blood pressure of 224/126 mm Hg captured objectively during an episode is a physiologic event, not an emotional one. The *pallor* is telling—anxiety and panic are typically accompanied by cutaneous flushing and warmth, whereas this man turns cold and white, a signature of intense peripheral vasoconstriction (though rare epinephrine-predominant tumors can cause flushing). And, decisively, panic disorder does not cause an 8-kg weight loss with a preserved appetite, new diabetes, an elevated hematocrit, or resistant hypertension. The failure of an adequate trial of sertraline to alter the spells is consistent with a nonpsychiatric cause. Panic disorder is a common misdiagnosis in patients who ultimately prove to have a catecholamine-secreting tumor; when "panic" is accompanied by objective hypertension and metabolic disturbance, the label must be challenged.

Thyrotoxicosis

Hyperthyroidism deserves explicit consideration, because it produces palpitations, fine tremor, heat intolerance, sweating, anxiety, and weight loss despite a normal or increased appetite—several of which this patient has. But the episodic, paroxysmal architecture of his symptoms is wrong for thyrotoxicosis, which produces a sustained hyperadrenergic state rather than discrete spells with near-normal intervals; he has no goiter, no ophthalmopathy, and no tachycardia at rest between episodes that would be expected. Most conclusively, the thyrotropin and free thyroxine are normal, which excludes overt thyrotoxicosis. This normal thyroid panel was a deliberate and necessary step, because thyrotoxicosis is a classic endocrine mimic of these spells.

Paroxysmal supraventricular tachycardia and other arrhythmias

A primary cardiac arrhythmia—paroxysmal supraventricular tachycardia in particular—can produce abrupt-onset palpitations with an abrupt offset, accompanied by anxiety, dyspnea, light-headedness, and, through sympathetic discharge, sweating and even polyuria afterward. This is worth weighing, because the palpitations are prominent. But an arrhythmia does not explain the pounding headache, the profound pallor, the sustained interictal hypertension, the weight loss, or the new diabetes, and the electrocardiogram captured during a spell showed *sinus* tachycardia, not a re-entrant supraventricular rhythm. The tachycardia here is the consequence of a systemic catecholamine surge, not its cause. The left ventricular hypertrophy on the electrocardiogram and the fourth heart sound point instead to the cardiac consequences of chronic hypertension and catecholamine excess.

Carcinoid syndrome and mast-cell disorders

Two neuroendocrine and immunologic conditions can produce paroxysmal "spells" and belong on this list. *Carcinoid syndrome* causes episodic flushing, diarrhea, palpitations, and wheezing from the secretion of serotonin and vasoactive peptides; *systemic mastocytosis and mast-cell activation syndrome* cause flushing, headache, palpitations, hypotension, and gastrointestinal symptoms from mediator release. Both are genuine causes of spells. But this patient's episodes are characterized by *pallor*, not flushing, and by *constipation*, not diarrhea; he has no wheeze, no urticaria pigmentosa, no pruritus, and no anaphylactoid hypotension. Carcinoid and mast-cell disease more often produce vasodilation and hypotension during attacks, the opposite of the severe vasoconstrictive hypertension documented here. If the catecholamine biochemistry had not been so revealing, a 24-hour urinary 5-hydroxyindoleacetic acid level and a serum tryptase level would have been reasonable, but the clinical phenotype already argues against both, and neither explains the sustained resistant hypertension.

Hypoglycemia and insulinoma

The adrenergic components of the spells—palpitations, diaphoresis, tremor, anxiety—are precisely the symptoms of a hypoglycemic catecholamine response, and an insulinoma can produce episodic neuroglycopenic and adrenergic attacks. But this patient is *hyperglycemic*, with a fasting glucose of 168 mg/dL and a glycated hemoglobin of 6.6%, which is the exact opposite of what an insulinoma would produce, and his spells bear no relationship to fasting. Hypoglycemia is therefore excluded; indeed, the new hyperglycemia is itself a clue that points toward the diagnosis, since catecholamine excess impairs insulin secretion and promotes glycogenolysis and lipolysis.

Drugs, toxins, and withdrawal states

A range of exogenous agents can reproduce this picture: cocaine and amphetamines and other sympathomimetics, the ingestion of tyramine-containing foods by a patient taking a monoamine-oxidase inhibitor, the abrupt withdrawal of clonidine or of a beta-blocker, and over-the-counter decongestants or stimulant-containing supplements. These must always be excluded by a careful history, because they are common and reversible. This patient is a nonsmoker who does not use illicit drugs or stimulants, takes no monoamine-oxidase inhibitor or clonidine, and uses no decongestants. His sertraline is a serotonin-reuptake inhibitor. Certain antidepressants—particularly tricyclics and serotonin–norepinephrine-reuptake inhibitors—can elevate measured metanephrines and must be accounted for when interpreting the biochemistry; that potential confounder was addressed in this case.

Autonomic dysfunction and baroreflex failure

Baroreflex failure, which may follow neck surgery or radiation or injury to the carotid baroreceptors, produces labile, paroxysmal hypertension with intervening normotension or hypotension and can superficially resemble

this case. Other autonomic disorders can also cause labile pressures. But this patient has had no neck surgery or irradiation, has no bulbar or autonomic neuropathic findings, and—again—has the metabolic and constitutional features and the family history that baroreflex failure does not explain. Baroreflex failure is the principal nontumoral cause of truly paroxysmal hypertension, but it does not unify the findings in this case.

Secondary hypertension: renovascular disease and primary aldosteronism

A 39-year-old man with resistant hypertension requires consideration of the structural and hormonal causes of secondary hypertension. *Renovascular hypertension*, from fibromuscular dysplasia in a patient of this age or from atherosclerotic disease, can cause severe and difficult-to-control hypertension; but he has no abdominal or flank bruit, normal renal function, and normal kidneys and renal arteries on cross-sectional imaging, and renovascular disease does not produce paroxysmal spells with pallor, sweating, and the classic headache–palpitation–diaphoresis triad. *Primary aldosteronism* is the most common endocrine cause of secondary hypertension and must be screened for, but it characteristically produces hypertension with hypokalemia and metabolic alkalosis, whereas this patient is normokalemic with a normal bicarbonate, and his aldosterone, renin, and aldosterone-to-renin ratio are all normal, excluding it. *Cushing syndrome* is argued against by the absence of any cushingoid stigmata and by the suppression of cortisol to below 1.8 µg/dL after low-dose dexamethasone—a result that also indicates that the adrenal mass is not a cortisol-secreting adenoma. Obstructive sleep apnea, coarctation, and renal parenchymal disease do not fit a paroxysmal symptom complex of this kind.

A nonfunctioning adrenal incidentaloma

Once an adrenal mass is identified, one must always ask whether it is an incidental, nonfunctioning adenoma that is unrelated to the clinical syndrome—a coincidence. Adrenal incidentalomas are common. But this lesion is decidedly not the bland, lipid-rich, low-attenuation adenoma that is usually incidental: it measures 4.8 cm, has an unenhanced attenuation of 38 Hounsfield units, shows no chemical-shift signal loss, displays a low contrast-washout pattern, is markedly hyperintense on T2-weighted imaging, and avidly concentrates MIBG. These features are not those of an incidental adenoma; they are the imaging signature of a catecholamine-secreting tumor, and the lesion is functionally and biochemically wedded to the clinical syndrome.

Convergence: a catecholamine-secreting tumor

The remaining syndrome is defined by catecholamine excess. The combination of episodic headache, palpitations, and diaphoresis carries a high specificity for a catecholamine-secreting tumor when it occurs together with hypertension, and the additional features here are all explicable by catecholamine excess: the *pallor* from alpha-adrenergic vasoconstriction; the *paroxysmal and sustained hypertension* from circulating norepinephrine and epinephrine; the *orthostatic hypotension* and the *elevated hematocrit* from chronic catecholamine-induced reduction of plasma volume and hemoconcentration; the *weight loss* and *hyperglycemia* from the calorigenic, lipolytic, glycogenolytic, and insulin-suppressing actions of catecholamines; the *constipation* from inhibition of gut motility; and the *provocation by exertion, bending, straining, and micturition*, which mechanically compress or stimulate the tumor and discharge its stored amines—the micturition trigger being a classic, if not invariably localizing, clue.

The biochemistry settles the matter. The single most sensitive test for this tumor is the measurement of fractionated metanephrines—the O-methylated metabolites of catecholamines—because the tumor produces these metabolites continuously within its own cells, so that they are elevated even between paroxysms, whereas the parent catecholamines fluctuate. This patient's plasma free metanephrine of 2.8 nmol/L and free normetanephrine of 9.6 nmol/L are elevated to roughly six-fold and more than ten-fold above their upper reference limits, respectively, and the 24-hour urinary fractionated metanephrines are concordantly and markedly elevated. The diagnosis of a catecholamine-secreting tumor is made biochemically, before and not after imaging. Modest elevations of metanephrines, below roughly twice the upper limit, can be false positives from medications, stress, posture, or illness; but elevations of this magnitude—several-fold above normal, obtained under controlled, supine, fasting conditions with interfering drugs excluded—are highly consistent with a catecholamine-secreting tumor and are rarely produced by anything else. The elevated chromogranin A, a marker of neuroendocrine secretory granules, corroborates the diagnosis.

The biochemical phenotype is also informative. Both metanephrine and normetanephrine are elevated, with a

substantial *metanephrine* component. Metanephrine is the metabolite of epinephrine, and epinephrine synthesis requires the enzyme phenylethanolamine N-methyltransferase, which is essentially confined to the adrenal medulla, where it is induced by the high local concentrations of cortisol. A tumor that produces appreciable epinephrine, and therefore metanephrine, is therefore almost certainly of *adrenal* origin rather than an extra-adrenal paraganglioma, and an adrenergic or mixed secretory phenotype is characteristic of tumors arising in certain hereditary contexts. The very high normetanephrine reflects the large norepinephrine output of a sizable tumor.

The imaging then performs its proper role in this disease: it localizes the source. The lesion is a 4.8-cm right adrenal mass with high unenhanced attenuation, no intracytoplasmic lipid, a low washout pattern, marked T2 hyperintensity with cystic and hemorrhagic change, and —most specifically—intense MIBG uptake confined to that single lesion with no extra-adrenal or metastatic foci. The constellation is that of a solitary adrenal catecholamine-secreting tumor, which is to say a pheochromocytoma.

The hereditary dimension and the "rule of ten"

The family history transforms the case from a likely sporadic tumor into a possible hereditary syndrome. The traditional teaching about this tumor is the "rule of tens"—that roughly 10% are bilateral, 10% extra-adrenal, 10% malignant, 10% occur in children, and 10% are familial. The first four of these remain useful approximations, but the last is now known to be a serious underestimate: with modern germline testing, approximately a third to 40% of pheochromocytomas and paragangliomas are associated with a hereditary mutation, the highest heritable fraction of any human tumor. The relevant syndromes are multiple endocrine neoplasia type 2, caused by activating mutations of the *RET* proto-oncogene; von Hippel–Lindau disease, caused by *VHL* mutations; neurofibromatosis type 1, caused by *NF1* mutations; and the familial paraganglioma syndromes caused by mutations in the succinate dehydrogenase subunit genes, *SDHB*, *SDHD*, and others.

This patient's family history points with some force toward multiple endocrine neoplasia type 2A. A father with hypertension and a surgically treated "thyroid tumor," followed by thyroid-hormone replacement and ongoing blood-test surveillance, is entirely compatible with medullary thyroid carcinoma—the defining tumor of this syndrome and one that frequently coexists with pheochromocytoma. The paternal relative with a tumor "near the kidney" may well have had an adrenal pheochromocytoma; lay descriptions readily conflate the adrenal gland with the adjacent kidney. The adrenergic, epinephrine-producing biochemical phenotype is itself typical of the pheochromocytomas seen in multiple endocrine neoplasia type 2. The patient's *normal serum calcium* does not argue against this syndrome, since primary hyperparathyroidism occurs in only a minority of affected persons, and the absence of mucosal neuromas and a marfanoid habitus argues against the more aggressive type 2B variant. Von Hippel–Lindau disease remains a consideration, but the kidneys were normal and the secretory phenotype is less purely noradrenergic than is typical of that syndrome; neurofibromatosis is excluded by the absence of café-au-lait macules, neurofibromas, and axillary freckling; and a succinate-dehydrogenase paraganglioma syndrome is less favored by the solitary adrenal, epinephrine-producing lesion, though it cannot be excluded without testing.

For all of these reasons, the clinical conclusion is that this man harbors a catecholamine-secreting tumor of the right adrenal medulla—a pheochromocytoma—most probably in the setting of an inherited cancer-predisposition syndrome, and most likely multiple endocrine neoplasia type 2A. The diagnostic and therapeutic procedure, once he had been adequately prepared pharmacologically, would be surgical removal of the adrenal tumor, with pathology expected to confirm a pheochromocytoma.

Clinical Diagnosis

Catecholamine-secreting tumor (pheochromocytoma) of the right adrenal gland, producing paroxysmal and sustained hypertension with secondary hyperglycemia and weight loss, most likely arising in the context of a hereditary pheochromocytoma–paraganglioma syndrome (probable multiple endocrine neoplasia type 2A).

Dr. Eleanor Whitfield's Diagnosis

Pheochromocytoma of the right adrenal gland, catecholamine-secreting, probably in the context of multiple endocrine neoplasia type 2A.

Pathological Discussion

Dr. Tomás Okafor (Pathology): The diagnostic procedure was a laparoscopic right adrenalectomy, performed after the preoperative pharmacologic preparation that Dr. Lindqvist will describe. The specimen consisted of the right adrenal gland bearing a well-circumscribed, encapsulated mass measuring 4.9 cm in greatest dimension and weighing 62 g. On sectioning, the cut surface was gray-tan to mahogany-brown and fleshy, with focal areas of cystic degeneration and recent and remote hemorrhage that accounted for the heterogeneity seen on imaging. A rim of compressed, golden-yellow adrenal cortex was identified at the periphery of the mass, and the lesion did not invade beyond the adrenal capsule. A classic, if informal, confirmatory observation was made at the grossing bench: immersion of a fresh fragment in a dilute potassium dichromate solution produced a brown discoloration, the chromaffin reaction from which these tumors take their name.

On microscopical examination of hematoxylin-and-eosin–stained sections, the tumor displayed the characteristic architecture of a pheochromocytoma. The neoplastic cells were arranged in well-defined nests and trabeculae—the so-called *Zellballen* pattern—separated by a delicate, richly vascular fibrous stroma. The cells were polygonal and moderately large, with abundant finely granular, basophilic-to-amphophilic cytoplasm corresponding to their dense content of catecholamine secretory granules, and with round-to-oval nuclei. Scattered nuclei showed the degenerative hyperchromasia and enlargement ("endocrine atypia") that is commonly seen in these tumors and that, in isolation, carries no prognostic significance. Mitotic figures were rare, and there was no atypical mitosis, no coagulative tumor necrosis, and no vascular or capsular invasion.

Immunohistochemical staining confirmed the diagnosis and established the lineage of the tumor. The neoplastic chief cells were diffusely and strongly positive for the neuroendocrine markers chromogranin A and synaptophysin, in keeping with their secretory phenotype and with the elevated serum chromogranin A. The transcription factor GATA3 was positive, consistent with chromaffin differentiation. A stain for S100 protein highlighted a delicate meshwork of *sustentacular cells* arranged at the periphery of the Zellballen nests—a feature highly characteristic of pheochromocytoma and paraganglioma. Critically, the tumor cells were *negative* for cytokeratins (AE1/AE3) and for the steroidogenic markers inhibin and melan-A, which excludes an adrenal cortical neoplasm and an epithelial (cortical or metastatic carcinoma) tumor, and confirms an origin from the adrenal *medulla*. The Ki-67 proliferation index was low, at approximately 2%.

I should comment on the assessment of malignant potential, because it is a point of frequent misunderstanding. The histologic features of this tumor were collated using the Pheochromocytoma of the Adrenal Gland Scaled Score, or PASS; the lesion scored below the threshold of 4 that has been proposed to separate tumors with a more aggressive potential, consistent with the indolent features described. However, it is essential to recognize that *no histologic scheme reliably distinguishes benign from malignant pheochromocytoma*. By current convention, malignancy in these tumors is defined not by histology but by the presence of metastatic disease at sites where chromaffin tissue is not normally found; the MIBG study showed no such disease here. For this reason, all of these tumors are regarded as having some malignant potential and require long-term surveillance, irrespective of their appearance.

Finally, because of the patient's age and family history, immunohistochemistry for the succinate dehydrogenase B subunit (SDHB) was performed as a screening tool for the heritable paraganglioma syndromes. SDHB staining was *retained* (granular cytoplasmic positivity was preserved), a result that argues against an underlying *SDHx* mutation but does not exclude all *SDHx*-related or other hereditary syndromes. On the basis of the young age at presentation, the suggestive family history, and the recognition that a large fraction of these tumors are heritable regardless of histology, the tumor and a blood sample were referred for germline genetic analysis. These findings—a circumscribed adrenal medullary tumor with a Zellballen architecture of chromogranin- and synaptophysin-positive chief cells, peripheral S100-positive sustentacular cells, GATA3 positivity, absent keratin and cortical-marker expression, and a low proliferation index—are diagnostic of a **pheochromocytoma of the right adrenal gland**.

Discussion of Management

Dr. Sandra Lindqvist (Endocrine Surgery): The management of a pheochromocytoma is a paradigm of the principle that careful preoperative preparation determines surgical safety. The central hazard is the intraoperative or peri-procedural catecholamine surge, which can precipitate a hypertensive crisis, arrhythmia, myocardial injury, or stroke; the corresponding goals of preparation are to block the effects of circulating catecholamines and to re-expand the chronically contracted intravascular volume.

The patient was therefore treated for fourteen days before surgery with an alpha-adrenergic receptor antagonist. Phenoxybenzamine, a noncompetitive, long-acting alpha-blocker, was started and titrated against his blood pressure and symptoms; selective alpha-1 antagonists such as doxazosin are an accepted alternative. After alpha-blockade was established and a degree of orthostatic blood-pressure reduction confirmed that it was adequate, he was encouraged to take a *liberal salt diet and increased fluids* to restore plasma volume, which counteracts the volume contraction responsible for his elevated hematocrit and orthostatic hypotension and prevents profound hypotension after the tumor is removed. Only *after* alpha-blockade was firmly in place was a beta-blocker added, to control the reflex tachycardia. The cardinal rule was observed: *beta-blockade must never be initiated before adequate alpha-blockade*. Blocking beta-2–mediated vasodilatation while alpha-mediated vasoconstriction is unopposed can precipitate a severe hypertensive crisis and acute heart failure. His sertraline had been discontinued during the evaluation, and his prior antihypertensive regimen was adjusted around the blockade. Standard preparation criteria—a seated blood pressure below about 130/80 mm Hg, mild orthostatic hypotension, and the absence of frequent ectopy or ischemic changes—were met before he was brought to the operating room.

Dr. Eleanor Whitfield (Endocrinology): Intraoperatively, with invasive arterial monitoring and adequate venous access, the anesthesia team managed the predictable hemodynamic phases. Manipulation of the tumor produced transient hypertensive surges, which were controlled with short-acting, titratable agents—sodium nitroprusside and phentolamine, with magnesium and esmolol available for surges and tachyarrhythmia. After the adrenal vein was ligated and the tumor's catecholamine output abruptly ceased, the expected fall in blood pressure occurred and was managed with the volume that preoperative repletion had restored and with brief vasopressor support. The laparoscopic right adrenalectomy was completed without complication.

The postoperative course illustrated the systemic reach of the disease. The patient was monitored for the recognized early complications—rebound hypotension and hypoglycemia, the latter arising from the sudden recovery of insulin secretion once catecholamine suppression is relieved—and his glucose was followed closely; he required no insulin and his glucose normalized. His blood pressure settled into the normotensive range without antihypertensive medication over the following days, his tremor and spells ceased entirely, and over subsequent weeks he regained weight. Plasma free metanephrines and normetanephrines were remeasured several weeks after surgery to confirm biochemical cure and had normalized completely, and the hematocrit drifted down into the mid-normal range as his plasma volume was restored.

In light of his age and family history, germline genetic testing was pursued, and it identified a pathogenic variant in the *RET* proto-oncogene at codon 634, establishing a diagnosis of multiple endocrine neoplasia type 2A. This result reframed the entire family: the father's "thyroid tumor" was almost certainly medullary thyroid carcinoma, and the affected relative's abdominal tumor was likely a pheochromocytoma. Accordingly, the patient was screened for the other components of the syndrome—serum calcitonin and carcinoembryonic antigen and thyroid ultrasonography for medullary thyroid carcinoma, and serum calcium and parathyroid hormone for hyperparathyroidism—and cascade genetic testing was offered to his first-degree relatives, including his children, with the recognition that gene carriers warrant consideration of risk-reducing thyroidectomy and lifelong biochemical surveillance. He was enrolled in indefinite annual biochemical monitoring for recurrent or metachronous (including contralateral) pheochromocytoma, which is essential because no histologic feature can exclude future malignant or recurrent disease.

The teaching points of this case are durable ones. First, the classic triad of episodic headache, palpitations, and diaphoresis, especially when joined to paroxysmal or res-

istant hypertension, must prompt evaluation for a catecholamine-secreting tumor; pheochromocytoma is often mislabeled as panic disorder, anxiety, or thyrotoxicosis, and the clue of *pallor* rather than flushing should redirect the clinician. Second, the diagnosis is made *biochemically before imaging*: fractionated metanephrines, plasma or urinary, are the most sensitive test, and marked elevations are essentially diagnostic, whereas imaging serves to localize a biochemically proven tumor and to assess for multifocal or metastatic disease. Third, preoperative preparation with *alpha-blockade followed only later by beta-blockade*, together with volume repletion, is what renders resection safe. And fourth, because a large fraction of these tumors are hereditary—far more than the traditional "10%"—every patient with a pheochromocytoma or paraganglioma should be considered for germline genetic testing, with attention to *RET*, *VHL*, *NF1*, and the *SDHx* genes, and to the screening and surveillance of affected kindreds.

Anatomical Diagnosis

Pheochromocytoma of the right adrenal gland (catecholamine-secreting paraganglioma of the adrenal medulla), arising in the setting of a germline *RET* pathogenic variant consistent with multiple endocrine neoplasia type 2A.